

Why Use a Virtual Environment in Python?

A virtual environment is an isolated Python environment that allows you to manage dependencies for different projects separately. This is crucial because:

- Different projects may require different versions of the same package.
- It prevents conflicts between dependencies across projects.
- It helps maintain a clean global Python environment.

Steps to Set Up a Virtual Environment in VS Code

1. Open the Integrated Terminal

Open the terminal by selecting **View > Terminal** or using the shortcut **Ctrl + `**.

2. Create a Virtual Environment

In the terminal, navigate to your project folder and run:

```
1 python -m venv myenv
2
```

Replace `myenv` with your desired virtual environment name.

3. Activate the Virtual Environment

Activate the environment with the following command:

• Windows:

```
1 myenv\Scripts\activate
2
```

• macOS/Linux:

```
1 source myenv/bin/activate
2
```

4. Select the Python Interpreter

If VS Code doesn't automatically detect the virtual environment, manually select it:

- (a) Open the Command Palette (**Ctrl + Shift + P**).
- (b) Type `Python: Select Interpreter`.
- (c) Choose the interpreter from the virtual environment.

5. Install Dependencies

With the environment activated, install packages using `pip`:

```
1 pip install package_name
2
```

6. Deactivate the Virtual Environment

When done, deactivate the environment with:

```
1 deactivate
2
```

Will VS Code Automatically Detect the Virtual Environment?

1. Automatic Detection

When you open a folder in VS Code that contains a virtual environment (e.g., in a subdirectory like `myenv`), VS Code should prompt you with a notification asking if you want to select that virtual environment as the Python interpreter for the workspace. If you accept, it will automatically activate that environment for you.

2. Manual Selection

If VS Code doesn't automatically detect the virtual environment, you can manually select it:

1. Press `Ctrl + Shift + P` to open the Command Palette.
2. Type `Python: Select Interpreter` and choose the interpreter located in your virtual environment's `Scripts` (Windows) or `bin` (macOS/Linux) directory.
3. Once selected, VS Code will use this virtual environment as the default interpreter for the workspace.